

Laith Assaf

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EDUCATION

Michigan State University, College of Engineering

East Lansing, MI

BS Computer Science, AI/ML | Dean's List x3

August 2022 - May 2027

- **Relevant Coursework:** Information Management & Cloud Computing (CSE 380), Software Design (CSE 335), Computer Systems (CSE 325)

EXPERIENCE

IT and Media Services Support Associate

May 2023 - Present

MSU College of Osteopathic Medicine

East Lansing, MI

- Configured and maintained 100+ classroom technology systems (projectors, audio systems, video conferencing) across 4 locations supporting **1,200+ students**, resolving 50-70 technical tickets weekly with 5-10 minute average response time
- Engineered systematic troubleshooting framework achieving **99.5% classroom uptime** and preventing \$50K annual losses through diagnostic methodology, resolving network connectivity, display output, audio routing, and Zoom conferencing issues within 5-minute SLA

AI Club - Workshop Coordinator

January 2025 - Present

Michigan State University

East Lansing, MI

- Developed and delivered 8+ technical workshops on **local LLM deployment** (Ollama, LM Studio), AI ethics, and ML fundamentals (**PyTorch**) to 500+ attendees across 15 sessions, achieving 90% retention and 4.8/5 satisfaction score

PROJECTS

SoundSense - Real-Time Audio Classification | *YAMNet, WebSocket* | **MHacks Winner**

October 2025

- **Won at MHacks 2025** (300+ participants, 24-hour hackathon) building real-time audio classification system using YAMNet CNN model on MemryX MXA2301 edge accelerators, achieving **95% accuracy** with <50ms inference latency across 521 AudioSet event classes for hearing-impaired users
- Architected **Flask backend** with **WebSocket streaming** for live hazard detection (emergency sirens, fire alarms, breaking glass), processing continuous audio streams with real-time classification updates and visual alerts delivered to web interface

Bonfire Platform & Ember Framework | *TypeScript, Node.js, Socket.io, Firebase, Next.js*

2025 – Present

- Designed and published **Ember**, an open-source TypeScript framework for multiplayer party games (@**bonfire-ember/core/server/client** npm packages), implementing phase state machine, WebSocket room orchestration, and player lifecycle into a 5-hook API enabling **80%+ code reuse** across new game builds
- Architected **multi-room orchestration system** on Express + **Socket.io** namespaces with **Firebase Realtime DB** state sync, hosting 4 concurrent games on a single **Railway** server with configurable disconnect strategies and TTL-based cleanup supporting 1,000+ concurrent rooms
- Built **Bonfire** consumer platform (**Next.js 16** on Vercel, Express on Railway, Firebase Auth) integrating 4 production party games with page-refresh session recovery and **380+ automated tests** across Ember packages

EXTRACURRICULAR ACTIVITIES

AI Club - Project Lead

January 2025 - May 2025

Michigan State University

East Lansing, MI

- Led 4-student team building Academic Performance Predictor using **GPT-4 API** and prompt engineering, achieving **80% prediction accuracy** across 50+ courses with OpenAI API and agile sprints

TECHNOLOGY SKILLS

Languages: Python, C++, C#, SQL, TypeScript

AI/ML: PyTorch, TensorFlow, OpenAI API, ML-Agents, YAMNet, OpenCV, Ollama

Web & Mobile: FastAPI, Flask, React, Node.js, Express, Socket.io, REST APIs, WebSockets, Firebase

Infrastructure & Tools: Unity, Git, Linux, Docker, SSH, AsyncIO, Railway, Vercel